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Isolation and Identification of Phytochemicals and Biological Activities of Hericium ernaceus and Their Contents in Hericium strains Using HPLC/UV Analysis

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ABSTRACT

Ethnopharmacological relevance

Hericium ernaceus has been traditionally used for the treatment of dyspepsia, gastric ulcer and enervation in traditional Chinese medicine for a long time.

Aim of the study

To examine the effect of Hericium strains on their ability to inhibit LPS and interferon-γ induced NO production in cell culture and the bioassay correlation of hericenone C, D, F, isolated from H. ernaceus.

Materials and methods

Hericenone C, D, F were isolated from H. ernaceus by open column chromatography and identified on the basis of spectroscopic analyses including ^{1}H-NMR, ^{13}C-NMR and MS. The amounts of hericenone C, D, and F in Hericium strains were determined by HPLC/UV analysis. In order to investigate the anti-inflammatory effect of Hericium strains extracts, RAW 264.7 cells were treated with 200 μg/mL of Hericium strains extracts for 48 h. Cell growth was assessed by MTT assay.

Results

Phytochemical constituents were isolated from H. ernaceus by open column chromatography. Their structures were elucidated as hericenones C, D, and F on the basis of spectroscopic analyses including ^{1}H-NMR, ^{13}C-NMR and MS. The amounts of

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hericenones C, D, and F in Hericium strains were determined by HPLC/UV analysis. Hericenones C, D, and F contents were highest in Norugungdenglee-2 (8.289 ± 0.593 mg/g), KFRI-1453 (4.657 ± 0.462 mg/g), and KFRI-1093 (5.408 ± 0.420 mg /g) strains, respectively. All Hericium strains extracts tested inhibited the lipopolysaccharide- and interferon-γ-induced inflammatory activity of RAW264.7 cells. The strain KFRI-1093 about 39.6 % reduced NO generation with compared to control.

Conclusion

We believe that the anti-inflammatory effect of KFRI-1093 was due to hericenone F content. Our results contribute towards validation of the traditional use, natural drugs and health supplements. And also, the developed simple, accurate and rapid LC method can be used determinate the content of hericenones from other Hericium strains.

Key words

Hericium ernaceus; quantitative analysis; anti-inflammatory; hericenone

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1. Introduction

Hericium erinaceum belonging to the family Hericiaceae is widely known as edible mushroom that grows on dead trunks of hard woods in Korea, Japan, China and Europe. H. erinaceum has been used as a cooking and medicine for the treatment of dyspepsia, gastric ulcer and enervation in traditional Chinese medicine for a long time (Traditional Chinese Medicine Editorial Committee, 1999; Inanaga, 2012; Miyazawa et al., 2012). H. erinaceum is known as the "yamabushi-take [yamabushi mushroom]" in Japan, while it is called the "monkey's head" mushroom in China and the "lion's mane" mushroom in the US (Inanaga, 2012).

Recently, the chemical constituents of H. erinaceum have been investigated for their interesting and significant bioactivities. The isolation and identification of constituents from this mushroom were hericenones A and B as cytotoxic constituents (Hirokazu et al., 1990; Yaoita et al., 2005a), hericenones C-E as stimulators of nerve growth factor (NGF)-synthesis (Kawagishi et al., 1991), 3,4-dihydro-5-methoxy-2-methyl-2-(4'-methyl-2'-oxo-3'-pentenyl)-9(7H)-oxo-2H-furo[3,4-h] benzeopyran (Yaoita et al., 2005b), hericenones F-H (Kawagishi et al., 1993), hericenone J as endoplasmic reticulum (ER) stress-suppressive compound (Uede et al., 2008), hericerinas pollen Growth Inhibitor (Kimura et al 1991), lectine (Kawagishi et al., 2001), fatty acid (Kawagishi et al., 1990, Shigefumi et al 1992), antitumor-active polysaccharides (Mizuno et al., 1992) other polysaccharide (Wang et al., 2004; Xu et al., 2010), glucan (Dong et al., 2006), laccase (Wang et al., 2004), erinacine D-I (Kawagishi et al 1996a; Kawagishi et al., 1996b), erinacine J-K (Kawagishi et al., 2006), erinacine P (Kenmoku et al., 2000), erinacine Q (Kenmoku et al., 2002), and ergosterol peroxide (Krzyczkowski et al., 2009). Especially,

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hericenones and the discovery of their physiological activity have focused attention on H. erinaceum as a health food ingredient that encourages nootropic action (Kawagishi et al., 1991; Kawagishi et al., 1992; Shimbo et al., 2005).

We screened extracts of various mushrooms and their mycelia for anti-inflammatory activity. Since the extract from the cultured mycelia of H. erinaceum showed potent activity, we tried to isolate the active principles from the fungus. This paper describes the isolation, structure elucidation, quantitative analysis and biological activities of the isolated compounds.

2. Material and methods

2.1. Microorganisms

Hericium erinaceum (Hericiaceae) was purchased from Pochun mushroom development Co.,Ltd at Pochun-gun in Gyunggi-Do province, Korea, in July 2010. A voucher specimen (SKKU-2010-07) of the mushroom was deposited in the herbarium of the School of Pharmacy, Sungkyunkwan University, Suwon, Korea. Hericium strains, which were deposited the voucher specimens in Rural Development Administration (RDA) and Korean Forest Research Institute (KFRI), korea, were provided by RDA and KFRI. The Characteristics and Origin of Hericium strains are shown in Tabel 1.

2.2. Reagents and instruments

Ethanol (EtOH), methanol (MeOH), n-hexane, chloroform ($CHCl_3$), dichloromethane (CH_2Cl_2), ethyl acetate (EtOAc), and n-butanol (n-BuOH) (SamChun Pure Chemical Co., Pyeongtaek, Korea) were used as the mobile phase in column chromatography. $CDCl_3$ (Sigma-Aldrich, St. Louis, MO) was used as the Nuclear magnetic resonance (NMR)

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solution. Electrospray ionization mass spectrometry (ESI-MS) was conducted with a Jeol JMS-700W (Tokyo, Japan) mass spectrometer. ^{1}H-, ^{13}C-NMR spectra were recorded with a Bruker AMX 500 NMR (Bremen, Germany) spectrometer using tetramethylsilane (TMS) as the internal standard. Chemical shifts are reported in parts per million (δ), and coupling constants (J) are expressed in Hertz. An Eyela rotary evaporator system (Tokyo, Japan) under reflux in vacuo was used for evaporation. Thin-layer chromatography (TLC) was conducted with Kiesel gel 60 F_{254} (Art. 5715, Merck Co., Darmstadt, Germany) plates (silica gel, 0.25 mm layer thickness), and compounds were visualized by spraying with 10% H_2SO_4 in MeOH, followed by heating to 100°C. Column chromatography was conducted with a Silica gel 60, RP-C_{18} silica gel (Merck, Germany, 70~230 mesh, and 230~400 mesh), and sephadex LH-20 column (bead size: 25-100 μ; Sigma-Aldrich Co., USA). Prep-HPLC was carried out on a Gilson 321 pump (France) with a fraction collector, a 1260 Quat pump VL, and a RI-101 (Shodex, Japan). Samples were separated using an Apollo Silica 5μ column (5 μm, 10 × 250 mm) and Econosil RP-18 10μ column (5 μm, 10 × 250 mm) at a flow rate of 8.0 mL/min. HPLC was carried out on an Agilent series 1260 separation system (Santa Clara, CA, USA), a 1260 Quat pump VL, and a 1260 Variable Wavelength Detector. Samples were separated using an INNO ® C18 (4.6 × 250 mm, 5 μm). The water and MeOH used in this research were of HPLC grade, and all other reagents were of analytical grade. RAW 264.7 cell was obtained from the Korea Cell Line Bank (KCLB, Seoul, Korea). Dulbecco's modified eagle medium (DMEM), fetal bovine serum (FBS), and penicillin/streptomycin were obtained from Welgene (Daegu, Korea). The lipopolysaccharide (LPS) used in this study was from Sigma Chemical Co. (MO, USA) and interferon-gamma (IFN-γ) was from Pepro Tech

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(NJ, USA). The Griess reagent, 3-(4,5-dimethylthiazol-2-yl)-2,3-diphenyl tetrazolium bromide (MTT), 3,3-tetramethylene glutaric acid (TMG), and dimethyl sulfoxide (DMSO) were obtained from Sigma Chemical Co. (MO, USA).

2.3. Cell culture

RAW 264.7 cells was cultured in DMEM containing 100 U/mL of penicillin/streptomycin and 10% FBS at 37 °C in a 5% CO_2 incubator. Cells were sub-cultured weekly with 0.05% trypsin-EDTA in phosphate buffered saline.

2.4. Cell viability assay

After confluence had been reached, the cells were plated at a density of 5 × 104 cells/well into 96 well plates, incubated for 2 hr and then treated with LPS (1 μg/mL) and IFN-γ (10 ng/mL). Samples were treated for 24 hr. After incubation, cell viability was determined using the MTT assay. MTT solution was added to each 96-well plate, the plates were incubated for 4 hr at 37 °C, and the medium containing MTT was removed. The incorporated formazan crystals in the viable cells were solubilized with 100 mL of DMSO and the absorbance of each well was read at 540 nm (Mosmann, 1983).

2.5. Measurement of nitrite

Nitric oxide (NO) production was assayed by measuring the accumulation of nitrite using a microplate assay method based on the Griess reaction (Sreejayan and Rao, 1997). RAW 264.7 cells were seeded in 96-well plates (5 × 104 cells/well) and LPS (1 μg/mL) and IFN-γ (10 ngmL) were added. After incubating the samples for 24 hr, 100 μL of culture supernatant was incubated with 100 μL of Griess reagent and the mixture was

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incubated at room temperature for 15 min. The optical density of the samples was measured at 540 nm using a microplate reader (Chiou et al., 1997).

2.6. Extraction and isolation

The half dried aerial parts of Hericium erinaceum (5.0 kg) were extracted with 80% MeOH two times at room temperature, and the extracts were combined and concentrated to obtain a brown residue (500 g). The residue was dissolved in water (10 L) and partitioned successively with n-hexane (3 × 10 L), CH_2Cl_2 (3 × 10 L), EtOAc (3 × 10 L), and n-BuOH (3 × 10 L) to give the n-hexane-soluble (63.3 g), CH_2Cl_2-soluble (4.5 g), EtOAc-soluble (5.0 g), and n-BuOH-soluble (17.5 g) fractions. The n-hexane extract was separated by silica gel column using hexane : EtOAc as the mobile phase (gradient: 50:1-1:1). Five fractions were obtained by combining those with similar R_f on TLC behavior (A-E). Fraction E (10g) was isolated using a reversed phase gel column, eluting with a solvent system of reversed phase C-18 MeOH (10 ~ 100%). According to TLC analysis, three crude fractions (fr. E1-E3) were collected Fr. E1 (1g) purified using a reversed phase C-18 (250 × 2.0 mm) column (gradient 40 ~ 70% MeOH), and purified further by preparative reverse-phase HPLC, using a solvent system of 60% MeOH, yielded compound 2 (38 mg). Fr. E3 (1 g) purified using a reversed phase C-18 (250 × 10 mm) column (gradient 50% ~ 70% MeOH), and purified further by preparative reverse-phase HPLC, using a solvent system of 50% MeOH, yielded compound 3 (36 mg).

The CH_2Cl_2 soluble fraction (4 g) was chromatographed on a silica gel column, eluting with a gradient solvent system of MC:MeOH (50:1-1:1) as the eluent to yield compound 1 (9.73 mg).

Compound 1 was obtained in the form of white powder. $C_{35}H_{50}O_6$; ESI-MS m/z: 589

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[M+Na]; ^{1}H-NMR (500 MHz, $CDCl_3$) δ: 12.36 (1H, s, OH), 10.08 (1H, s, CHO), 6.51 (1H, s, H-6), 6.07 (1H, s, H-6'), 5.30 (3H, s, H-2', H-1' CH_2), 3.89 (3H, s, OCH_3), 3.37 (2H, d, J = 7.2 Hz, H-1), 2.99 (1H, s, H-4'), 2.34 (2H, dd, J = 7.7, 7.3 Hz, H-2), 2.12 (3H, s, H-7' CH_3), 1.84 (3H, s, H-8'), 1.78 (3H, s, H-3' CH_3), 0.88 (3H, t, J = 7.0 Hz, terminal CH_3); ^{13}C-NMR (125 MHz, $CDCl_3$) δ: 199.7 (C-5'), 193.3 (CHO), 173.4 (C-1), 163.7 (C-5), 163.2 (C-3), 155.6 (C-7'), 138.9 (C-1), 130.5 (C-3'), 126.4 (C-2'), 123.0 (C-6'), 117.5 (C-4), 113.1 (C-2), 105.7 (C-6), 63.1 (C-1' CH_2), 56.1 (OCH_3), 55.7 (C- 4'), 34.4 (C-2''), 27.9 (C-8'), 21.8 (C-1'), 20.9 (C-9' CH_3), 16.6 (C-10' CH_3), 14.3 (C-16'' or 18''), 32.1, 29.7, 29.9, 29.8, 29.7, 29.6, 29.4, 29.3, 25.1, 22.9 (C-3 to C-15 or C-17).

Compound 2 was obtained in the form of yellow powder. $C_{37}H_{58}O_6$; ESI-MS m/z: 621 $[M+Na]^+$; ^{1}H-NMR (500MHz, $CDCl_3$), δ 12.37 (1H, s, OH), 10.10 (1H, s, CHO), 6.52 (1H, s, H-6'), 6.08 (1H, s, H-6), 5.32 (2H, s, H-1', CH_2), 5.32 (1H, t, J = 7.0 Hz, H-2), 3.91 (3H, s, OCH_3), 3.48 (2H, d, J = 7.0 Hz, H-1), 2.99 (2H, s, H-4'), 2.32 (2H, dd, J = 8.0, 7.5 Hz, H-2''), 2.11 (3H, s, H-9'), 1.84 (3H, s, H-8'), 1.77 (3H, s, H-10'), 1.60 (2H, m, H-3''), 1.24 (2H, m, H-7'', H-15''), 0.86 (3H, t, J = 7.0 Hz, H-18) ^{13}C-NMR (125MHz, $CDCl_3$), δ 199.6 (C-5'), 193.2 (CHO), 173.3 (C-1''), 163.6 (C-5), 163.1 (C-3), 155.5 (C-7'), 138.9 (C-1), 130.5 (C-3'), 126.4 (C-2), 123.0 (C-6'), 117.5 (C-4), 113.1 (C-2), 105.7 (C-6), 63.0 (C-1' CH_2), 56.1 (OCH_3), 55.7 (C-4'), 34.4 (C-2''), 23.1 (C-16''), 27.8 (C-8'), 25.0 (C-3''), 22.8 (C-17''), 21.8 (C-1'), 20.8 (C-9'), 16.6 (C-10'), 14.3 (C-18''), 29.87, 29.85, 29.83, 29.78, 29.64, 29.56, 29.42, 29.32 (C-4'' to 15'').

Compound 3 was obtained in the form of yellow powder. $C_{35}H_{54}O_6$; ESI-MS m/z: 593 $[M+Na]^+$; ^{1}H-NMR (500MHz, $CDCl_3$), δ10.40 (1H, s, CHO), 6.53 (1H, s, H-6), 6.05 (1H, s, H-3'), 5.50 (2H, s, H-7' CH_2), 3.87 (3H, s, OCH_3), 2.83 (1H, d, J= 14.5 Hz, H-1'a), 2.66

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(1H, d, J= 13.5 Hz, H-1'b), 2.64 (2H, dd, J= 5.5, 7.0Hz, H-4), 2.40 (2H, t, J=7.5Hz,H-2''), 2.13 (3H, s, H-6'), 2.03 (1H, m, H-3a), 1.92 (1H, m, H-3b),1.85 (3H, s, H-5'), 1.66 (2H, m, H-3''), 1.43 (3H, s, H-2a), 0.87 (3H, t, J= 7.0 Hz, H-16'') ^{13}C-NMR (125MHz, $CDCl_3$), δ 198.0 (C-2'), 190.4 (CHO), 173.4 (C-1''), 162.1 (C-5), 158.4 (C-8a), 156.5 (C-4'), 139.9 (C-7), 125.1 (C-3'), 116.0 (C-8), 109.4 (C-4a), 101.0 (C-6), 77.1 (C-2), 64.7 (C-7' CH_2), 55.7 (OCH_3), 52.6 (C-1'), 34.6 (C-2''), 30.2 (C-3), 27.9 (C-5'), 25.3 (C-3''), 24.7 (C-2a), 21.0 (C-6'), 16.6 (C-4), 14.3 (C-16''), 32.1, 29.8, 29.8, 29.8, 29.6, 29.5, 29.5, 29.4, 29.3, 22.8 (C-4'' to 15'').

2.7. Preparation of standard compounds

Hericenone C, hericenone D, and hericenone F were isolated from Hericium erinaceum (Fig. 1). To quantify the amount of hericenone C, hericenone D, and hericenone F 1 mg of each standard compounds were dissolved in 100 μL $CHCl_3$. The resultant solutions were filtered with a 0.45-μm filter. The resulting solution was used for HPLC analysis.

2.8. Procedure and data processing

Air-dried, milled powder of Hericenone strains were extracted with 70 % EtOH (300 mL × 3) room temperature, and the extracts were combined and evaporated to afford a brown residues. The residue was dissolved in 500 μL of $CHCl_3$ and filtered with a 0.45 μm filter (Whatman,UK). The resulting solution was used for HPLC analysis. The each individual 70 % EtOH extract (1.0 mg) was dissolved in DMSO (5 mL) for anti-inflammatory analysis.

2.9. HPLC condition

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The HPLC separation of hericenone C, hericenone D, and hericenone F for qualitative and quantitative analysis was performed using a reverse phase system. A reverse phase column (INNO® C18 (4.6 × 250 mm, 5 μm) used, with a mobile phase consisting of Methanol and Water. The gradient solvent system was 20:80 initially, and increased in linear gradients to 50:50 for 10 min, then to 0:100 for 10 min, and finally to 20 : 80 for 5 min. UV detection was conducted at 295 nm. The injection volume was 10 μL and the flow rate was 0.8 mL/min. All injections were performed in triplicate.

2.10. Calibration

A stock solution (1 mg/1 mL) of five compounds, derivatives isolated H. erinaceum was prepared in $CHCl_3$. The contents of the analytics were determined from the corresponding calibration curves. The calibration functions of flavone derivatives are calculated with peak area (Y), concentration (X, μg/10 μL), and mean values (n = 3) ± standard deviation.

2.11. Limit of detection (LOD) and limit of quantification (LOQ)

Validation of the HPLC method for compounds 1, 2 and 3 as standard compounds was determined by LOD and LOQ. Method linearity was established by triplicate injections in the range of 0.015-1.0 mg/ml. Seven calibration solutions were injected in triplicate, and five replicate analyses of the calibration solutions were performed. Calibration curves were constructed by linear regression of the peak area-ratios (Y) of compounds 1, 2 and 3 versus concentration (X) in mg/ml. The relative standard deviation was used as a measure of repeatability. The percent recoveries were evaluated by calculating the ratio of amount detected versus amount added. LOD and LOQ values were determined

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separately at signal to noise ratios (S/N) of 3 and 10, respectively.

3. Results and discussion

3.1. Identification of compounds 1-3

Compound 1 was obtained as a colorless white powder. The molecular formula of 1 was deduced to be $C_{35}H_{50}O_6$ by positive mode electrospray ionization mass spectrometry (ESI-MS) data at m/z 593 $[M+Na]^+$. The IR spectrum of 1 showed absorption bands at 3356 and 1635 cm-1 ascribable to hydroxyl and carbonyl functions, respectively. The ^{1}H-NMR spectrum of 1 displayed signals for four methyl groups (δ 2.12, 1.84, 1.78 and 0.88), one methoxy group (δ 3.89), one aromatic proton (δ 6.51), two methylene proton signals (δ 3.37 and 2.99), and oxymethylene proton signals (δ 5.30). In the ^{13}C-NMR spectrum, 35 carbon signals appeared besides those of the fatty acid unit, which included one ketone (δ 199.7), six aromatic carbons (δ 163.7, 163.2, 138.9, 117.5, 113.1 and 105.7), four methyl carbons (δ 27.8, 20.9, 16.6 and 14.3), two methylene carbon (δ 63.1, 55.7), one methoxy group (δ 56.1), four olefinic carbons (δ 155.6, 130.5, 126.4 and 123.0). The NMR data were very similar to those of hericenone C, which was isolated from H. erinaceum. Based on further comparison of the NMR data with those in the literature (Lee et al., 2008), the structure of 1 was determined to be hericenone C.

Compound 2 was obtained as a colorless yellow powder. The molecular formula of 2 was deduced to be $C_{37}H_{58}O_6$ by positive mode electrospray ionization mass spectrometry (ESI-MS) data at m/z 621 $[M+Na]^+$. The IR spectrum of 2 showed absorption bands at 3356 and 1635 cm-1 ascribable to hydroxyl and carbonyl functions, respectively. The ^{1}H-

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NMR spectrum of 2 displayed signals for four methyl groups (δ 2.11, 1.84, 1.77 and 0.86), one methoxy group (δ 3.91), one aromatic proton (δ 6.52), two methylene proton signals (δ 3.48 and 2.99), and oxymethylene proton signals (δ 5.32). In the ^{13}C-NMR spectrum, 37 carbon signals appeared besides those of the fatty acid unit, which included one ketone (δ 199.6), six aromatic carbons (δ 163.6, 163.1, 138.9, 117.5, 113.1 and 105.7), four methyl carbons (δ 27.8, 20.8, 16.6 and 14.3), two methylene carbon (δ 63.0, 55.7), one methoxy group (δ 56.1), four olefinic carbons (δ 155.5, 130.5, 126.4 and 123.0). The NMR data were very similar to those of hericenone D, which was isolated from H. erinaceum. Based on further comparison of the NMR data with those in the literature (Hirokazu et al., 1999), the structure of 2 was determined to be hericenone D.

Compound 3 was obtained as a colorless gum. The molecular formula of 3 was deduced to be $C_{35}H_{54}O_6$ by positive mode electrospray ionization mass spectrometry (ESI-MS) data at m/z 593 $[M+Na]^+$. The IR spectrum of 3 showed absorption bands at 3366 and 1669 cm-1 ascribable to hydroxyl and carbonyl functions, respectively. The ^{1}H-NMR spectrum of 3 displayed signals for four methyl groups (δ 2.13, 1.85, 1.43 and 0.87), one methoxy group (δ 3.87), one aromatic proton (δ 6.53), three methylene proton signals (δ 2.83, 2.66, and 2.64), oxymethylene proton signal (δ 5.50). In the ^{13}C-NMR spectrum, 35 carbon signals appeared besides those of the fatty acid unit, which included one ketone (δ 198.0), six aromatic carbons (δ 162.1, 158.4, 139.9, 116.0, 109.4 and 101.0), four methyl carbons (δ 27.9, 24.7, 21.0 and 14.3), one methylene carbon (δ 52.6, 30.2 and 16.6), oxymethylene carbon (δ 64.7), one methoxy group (δ 55.7), two olefinic carbons (δ 156.5 and 125.1). The NMR data were very similar to those of hericenone F, which was isolated from H. erinaceum. Based on further comparison of the NMR data

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with those in the literature (chromans, 1993), the structure of 3 was determined to be hericenone F.

3.2. Anti-inflammatory activity

LPS is a cell wall component of all Gram-negative bacteria. LPS-induced activation of macrophages led to production of bioactive lipids, ROS, and in particular, inflammatory cytokines (Beutler and Rietschel, 2003). LPS and IFN-γ can synergistically stimulate cells to produce a large amount of NO (Nathan, 1992). The synthesis of NO, a mediator of inflammatory function, was greatly increased when RAW 264.7 cells were co-stimulated with LPS and IFN-γ. Increased NO production is a common phenomenon that occurs in LPS/IFN-γ-stimulated macrophages and is used as an indicator of inflammatory response. As shown in Table 2, when Hericium strains extracts were added to LPS and IFN-γ-stimulated RAW 264.7 cells, NO production was decreased about 39.6 %, as compared to the control group. To investigate the anti-inflammatory effect of Hericium strains extracts, RAW 264.7 cells were treated with 200 μg/mL of Hericium strains extracts for 48 h. Cell growth was assessed by MTT assay and the result is shown in Table 2. NO production from LPS/IFN-γ-activated RAW 264.7 cells was inhibited by Hericium strains extracts (KFRI-582, -842, -1093, -1094, -1095, -1452, and -1951), which showed less than 90% of NO generation at a non-toxic concentration. KFRI-1093 significantly inhibited the production LPS/IFN-γ-induced NO at the concentration of 200 μg/mL. Therefore, KFRI-1093 might have the potential to be used as therapeutic agents for preventing inflammatory diseases.

3.3. Quantitative analysis of herinenone C, D, and F in Hericium strains

The KFRI-1093 attracts a considerable attention for their significant biological

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activities including the anti-inflammatory activity. In order to determine the part-specific preponderance of KFRI-1093 in Hericium strains, HPLC analysis was conducted. The gradient elution profile was optimized to obtain the highest resolution of hericenones and the shortest time of analysis.

Three herinenones content of Hericium strains were determined by HPLC (Fig. 2). The calibration equation and retention times were shown in Table 3 and 4. The retention time of the expected peak of each hericenone in the Hericium strains were the same as that of the standard compounds. The most content of herinenone C, D, and F in the Hericium strains were highest in Norugungdenglee-2 (8.289 ± 0.593 mg/g), KFRI-1453 (4.657 ± 0.462 mg/g), and -1093 (5.408 ± 0.420 mg /g), respectively. As a result, the herinenone F content of the KFRI-1093 is higher than that of others.

The LOD and LOQ under the present chromatographic conditions were determined at signal to-noise ratios (S/N) of 3 and 10, respectively. The LOD and LOQ of hericenone C (1), hericenone D (2) and hericenone F (3) were 0.019 - 0.057, 0.042-0.139 and 0.022 - 0.074 mg/ml, respectively (Table 5).

4. Conclusion

It is very important herinenone F, we suggest that the anti-inflammatory effect of KFRI-1093 by herinenone F content. The presence of herinenone F in KFRI-1093 is especially considerable in by-products of agricultural crop for increasing the amounts of natural drugs and health supplements. A simple, accurate and rapid LC method was developed as the first report of an LC simultaneous determination of three main hericenones in Hericium strains. The results indicate that samples from different strains

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of the Hericium show a specific and similar LC chromatogram. This method can be used to determinate the content of hericenones from other Hericium strains. The presence of many kinds of herinenones in H. erinaceum suggests good biological potentials for this natural product with respect to its wide distribution in the world.

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Table 1

Lists of Hericium erinaceum isolates used in this experiment.

Strains	Characteristics	Origin
Norugungdenglee 1	local variety	RDA[1)]
Norugungdenglee 2	local variety	RDA
KFRI-349	Wild type	KFRI[2)]
KFRI-507	Wild type	KFRI
KFRI-508	Wild type	KFRI
KFRI-509	Wild type	KFRI
KFRI-582	Wild type	KFRI
KFRI-842	Wild type	KFRI
KFRI-843	Wild type	KFRI
KFR-844	Wild type	KFRI
KFRI-1090	Wild type	KFRI
KFRI-1091	Wild type	KFRI
KFRI-1093	Wild type	KFRI
KFRI-1094	Wild type	KFRI
KFRI-1095	Wild type	KFRI
KFRI-1452	Wild type	KFRI
KFRI-1453	Wild type	KFRI
KFRI-1622	Wild type	KFRI
KFRI-1623	Wild type	KFRI
KFRI-1950	Wild type	KFRI
KFRI-1951	Wild type	KFRI

[1)]Rural Development Administration(RDA), [2)]Korean forest research Institute (KFRI)

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Table 2

Anti-inflammatory activites of Hericium strains.

strain	NO generation (%)	Cell viability
Norugungdenglee 1	73.973 ± 6.443	74.339 ± 1.521
Norugungdenglee 2	72.596 ± 5.869	75.227 ± 3.225
KFRI-349	69.658 ± 7.786	77.190 ± 2.780
KFRI-507	71.620 ± 6.601	81.552 ± 2.637
KFRI-508	73.096 ± 6.855	77.492 ± 1.571
KFRI-509	72.114 ± 6.420	89.275 ± 6.729
KFRI-582	72.883 ± 5.986	91.295 ± 3.103
KFRI-842	74.419 ± 5.851	90.370 ± 4.000
KFRI-843	74.928 ± 2.792	87.255 ± 2.267
KFR-844	72.982 ± 2.397	87.160 ± 1.424
KFRI-1090	70.468 ± 3.150	85.668 ± 3.469
KFRI-1091	73.709 ± 3.655	89.974 ± 4.050
KFRI-1093	60.458 ± 2.700	97.961 ± 7.451
KFRI-1094	72.008 ± 2.995	96.677 ± 1.440
KFRI-1095	71.210 ± 2.826	92.938 ± 2.739
KFRI-1452	72.923 ± 2.404	90.332 ± 1.597
KFRI-1453	72.786 ± 5.977	85.857 ± 3.460
KFRI-1622	69.654 ± 7.967	85.159 ± 2.509
KFRI-1623	70.880 ± 5.867	83.591 ± 3.104
KFRI-1950	72.494 ± 7.183	87.481 ± 1.516
KFRI-1951	71.869 ± 7.014	90.238 ± 2.274
Control	100.000 ± 15.165	100.000 ± 1.956
AMT*	58.03	

*AMT (10 μg/ml) was used as a positive control.

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Table 3

The phytochemical contents in Hericium strains.

strain	Content (mg/g)		
	1	2	3
Norugungdenglee 1	6.582 ± 0.336	0.743 ± 0.037	0.394 ± 0.043
Norugungdenglee 2	8.289 ± 0.593	1.643 ± 0.060	0.158 ± 0.028
KFRI-349	1.153 ± 0.252	2.817 ± 0.218	trace
KFRI-507	0.799 ± 0.112	1.557 ± 0.132	trace
KFRI-508	0.238 ± 0.050	2.733 ± 0.225	trace
KFRI-509	2.071 ± 0.178	0.172 ± 0.026	trace
KFRI-582	5.951 ± 0.536	0.712 ± 0.138	0.691 ± 0.084
KFRI-842	0.177 ± 0.025	2.344 ± 0.262	trace
KFRI-843	1.110 ± 0.231	2.047 ± 0.139	trace
KFR-844	trace	0.181 ± 0.012	3.971 ± 0.164
KFRI-1090	7.444 ± 0.616	0.242 ± 0.014	1.128 ± 0.133
KFRI-1091	2.692 ± 0.437	0.078 ± 0.014	trace
KFRI-1093	trace	0.202 ± 0.011	5.408 ± 0.420
KFRI-1094	trace	0.162 ± 0.016	0.662 ± 0.107
KFRI-1095	1.631 ± 0.127	trace	trace
KFRI-1452	1.500 ± 0.178	4.634 ± 0.208	trace
KFRI-1453	2.423 ± 0.098	4.657 ± 0.462	trace
KFRI-1622	2.313 ± 0.353	3.456 ± 0.255	0.293 ± 0.045
KFRI-1623	trace	3.241 ± 0.180	trace
KFRI-1950	3.814 ± 0.285	4.395 ± 0.303	trace
KFRI-1951	1.992 ± 0.175	2.354 ± 0.249	trace

Data are represented as the mean ± S.D. (n = 3) in mg/g of the dried samples.

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Table 4

Linearity of standard curves of compounds 1-3.

Compound	t_R	Calibration equation [a]	Correlation factor, r^2 [b]
1	30.4	Y = 515.6X + 0.8958	1
2	28.4	Y = 478.81X + 0.9161	1
3	31.9	Y = 636.47X + 0.8983	1

[a] Y = peak area, X = concentration of standards (mg/mL).

[b] r^2 = correlation coefficient for three data points in the calibration curves (n = 3).

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Table 5

LOD and LOQ values of compounds 1-3.

Compound	Regression equation	r^2	Linear range (mg/ml)	LOD (mg/ml)	LOQ (mg/ml)
1	Y = 34.053X + 0.0351	1	0.0062-0.1	0.019	0.057
2	Y = 15.69X + 0.2986	1	0.0062-1.0	0.042	0.139
3	Y = 31.053X – 0.0398	1	0.0062-0.1	0.022	0.074

Y = Peak area, X = Concentration of standard (mg/ml)

r^2 = Correlation coefficient for three data points from calibration curve

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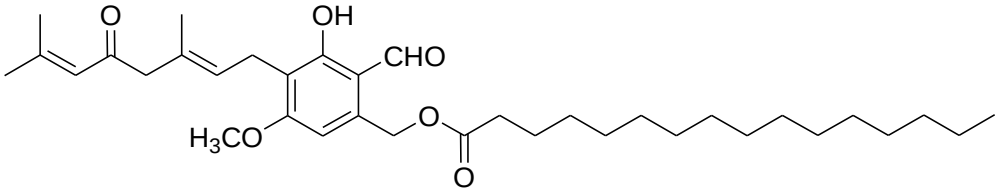

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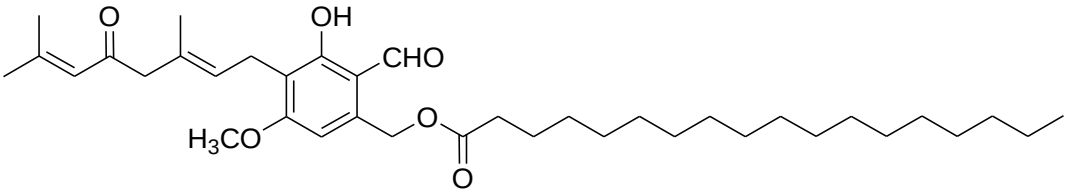

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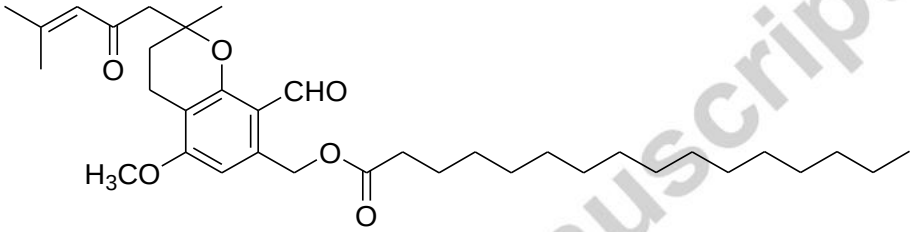

3

Fig. 1. Structure of compounds 1-3.

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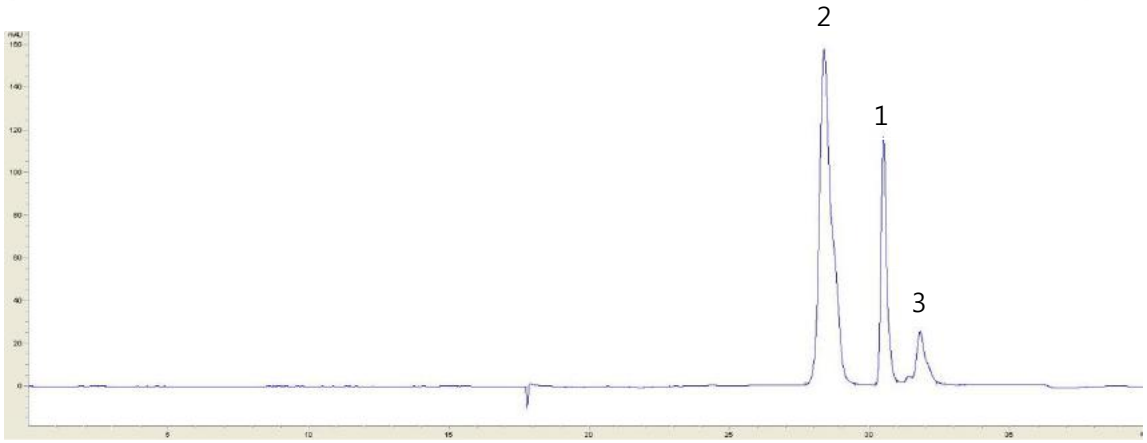
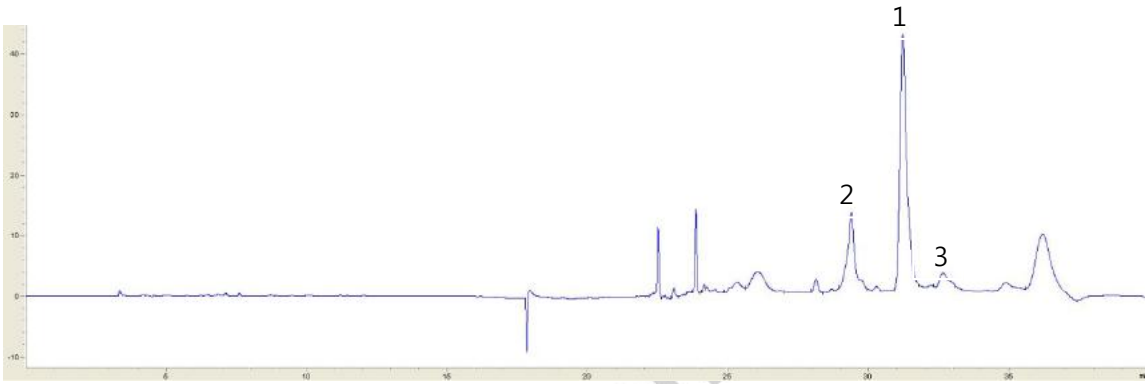


a



b

Fig. 2. LC chromatograms of the three compounds (a) and KFRI-1093 (b). Peaks 1 = hericenone-C; 2 = hericenone-D; 3 = hericenone-F

Graphical abstract

- ✓ The anti-inflammatory of Hericium strains

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Hericium ernaceus

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